1 What is use of Asymptotic Notation

- removes details that depend on architecture, language, compiler, etc.

- Still percise to identify different algorithms

* suppress constant factors and lower order terms *

Big O formal def:

If there exists positive constants c and n_0 such that $0 \leq T(n) \leq cg(n)$ for all

Game theory view:

To prove $T(n) = O(f(n))$ you commit fixed c and n_0 your opponent picks any $n \geq n_0$.

If $T(n) \leq c\, f(n)$ } you always win
If $T(n) = O(f(n))$ }

ex 21-4

2.1 c ✓ 2.2 c ✓ 2.3 d ✓ 2.4 d ✓

key formal result

If $T(n) = a_k n^k + \ldots + a_1 n + a_0$ then $T(n) = O(n^k)$

$n^k \neq O(n^{k-1})$

Big Omega notation (lower bound)

$T(n) = \Omega(f(n))$ iff there exists positive constants c, n_0 such that $T(n) \geq c \cdot f(n)$ for all $n \geq n_0$.

Big theta Notation (tight bound)

$T(n) = \Theta(f(n))$ iff $T(n) = O(f(n))$ and $T(n) = \Omega(f(n))$

And there exists $c_1, c_2, n_0 > 0$ such that $c_1 \cdot f(n) \leq T(n) \leq c_2$ for all $n \geq n_0$.

2.5 b, c, d

Little O Notation (strict upper bound)

$T(n) = o(f(n))$ iff for every constant $c > 0$ there exists n_0 such that $T(n) \leq c \cdot f(n)$ for all $n \geq n_0$

In big O you pick c once

In little O opponent can pick c and it can be very small

Adding constant to exponent: $2^{n+10} = O(2^n)$

Multiplying constant to exponent: $2^{10n} \neq O(2^n)$ since $2^{an} \leq c$ is false

Max vs Sum

$\max\{f(n), g(n)\} = \Theta(f(n)+g(n))$ since $\frac{1}{2}(f+g) \leq \max\{f,g\} \leq f+g$

so $c_1 = \frac{1}{2}$ $c_2 = 1$

2a $f(n) = 2$ $g(n) = 5$

Yes both are constants $c=1$ and $n_0 = 1 \Rightarrow 2 \leq 1 \cdot 5$ for all $n \geq 1$ ✓

2b $f(n) = 5n$, $g(n) = 7$

No, $f(n) = 5n$ has no bound and $g(n)$ is constant

2c $f(n) = 3n^2 + 7n$ $g(n)$ $2n^2$

yes for $n \geq 7$: $7n \leq n^2$ so $3n^2 + 7n \leq 2n^2 + n^2$

$= 4n^2 = 2 \cdot 2n^2 = 2g(n)$

Set $c=2$ and $n_0 = 7$ $f(n) = O(n^2) = O(g(n))$ ✓

---

3 $10n^2 = O(n^3)$

$10n^2 \leq c \cdot n^3$ for $n \geq n_0$

$10 \leq cn$ $c=1$ $n_0 = 10$ for all $n \geq 10$

$10n^2 \leq 1 \cdot n \cdot n^2 = n^3$ ✓